SAEON Rainfall Monitoring Sprint Backlog

ESP32 prototype with cellular MQTT and a live dashboard

This is the 13-week project plan for SAEON, running from 28 July to 23 October 2026 with work scheduled Tuesday to Friday. Each weekly sprint matches the product backlog and identifies its daily tasks, lead responsibilities, acceptance checks and current status.

The prototype counts TR-525M rain-gauge tips, reads an SHT31 sensor through the ESP32 WROOM-32, and sends readings through the SIM7000G to HiveMQ Cloud. Firebase supports the dashboard, historical data and CSV export. Solar/battery integration and station health are included; final deployment and field validation remain In Progress.

Team responsibilities

Team member	Charter role	Backlog focus
Redah Gamieldien	Project Leader	Requirements, hardware integration and final sign-off
Angelo Adams	Frontend Developer and Designer	Dashboard design and build, and making it easy to use
Qaasim Isaacs	Backend Developer and Tester	MQTT data handling, integration and test evidence
Ethan Williams	Analyst Researcher and Writer	Requirements, research and writing things up
Rameez Karriem	Agile Lead and Presenter	Planning, tracking tasks, running reviews and presenting

Working scope

The selected controller is ESP32 WROOM-32 and the selected temperature and humidity sensor is SHT31. LilyGO and DHT11 were earlier or exploratory options. The final communication route is SIM7000G cellular MQTT through HiveMQ Cloud, with Firebase for stored readings. Earlier HTTP-based experiments are development history, not the final architecture.

The current scope includes solar charging, battery monitoring, station diagnostics, historical data, CSV export, authenticated dashboard access and permanent hardware assembly. Extra pressure, wind and UV sensors, GPS and over-the-air firmware updates remain future enhancements. Final field acceptance depends on site access, power, protection and completed validation.

Planning basis

The 13 weekly sprints follow the current Trello-derived product backlog. Named leads coordinate tasks with the team; Ethan supports research and documentation across the project. Historical task estimates are not transferred to the revised schedule because the work breakdown has changed. Revised estimates and actual hours are not recorded.

How we manage the backlog

There is one sprint backlog for each of the 13 project weeks. Statuses match the product backlog: 43 daily task cards are Completed and six deployment or field-validation cards are In Progress. These are project-owner updates, not independently verified test results.

Weekly workflow

We plan weekly, check team capacity and use the agreed progress check-ins. Trello tracks tasks; GitHub stores code, documentation and evidence. Daily logbook entries are individual work notes and do not imply a formal stand-up meeting every day.

The shared task columns are Product Backlog, In Progress and Completed. Record any blocker, its owner and next action in the relevant task. Do not mark field work Completed until its acceptance evidence is available.

At the end of each sprint we demo what we built and write down any feedback from the lecturer or SAEON. Then we hold a short retrospective and pick one improvement to carry into the next sprint. Anything unfinished goes back to the product backlog so we can reprioritise it.

Definition of Done

- The task is finished and passes its acceptance checks, with the evidence linked on GitHub.
- Someone else on the team has reviewed it and the relevant tests pass. Any defects or limits are written down.
- The code, wiring diagrams, setup instructions and any other affected documents are updated.
- The work meets the agreed rainfall, MQTT and dashboard requirements for that item.
- The project leader signs it off and it is ready to demo or hand in.

Sprint plan at a glance

Week	Selected item	Focus	Est. hours
1	PB01	Project Initiation & System Architecture	Not recorded
2	PB02	Core Firmware & Sensor Interfacing	Not recorded
3	PB03	HiveMQ Cloud Broker Setup	Not recorded
4	PB04	Cellular Communication (SIM7000G)	Not recorded
5	PB05	SAEON Dashboard Base Application	Not recorded
6	PB06	Power Supply & Solar Integration	Not recorded
7	PB07	Advanced UI & Health Diagnostics	Not recorded
8	PB08	Hardware Finalization & PCB Soldering	Not recorded

Week	Selected item	Focus	Est. hours
9	PB09	Pre-Deployment Auditing & Calibration	Not recorded
10	PB10	48-Hour Field Stress Test	Not recorded
11	PB11	Presentation Formatting & Content	Not recorded
12	PB12	Rehearsal & Submission Prep	Not recorded
13	PB13	Project Buffer & Final Review	Not recorded

The schedule below uses 13 workweeks. Earlier eight-week estimates have been superseded; revised person-hour estimates are not recorded. Calendar end dates are planning periods, not proof that all work finished within them.

Product backlog

PB01 to PB13 map one-to-one to the 13 sprint groups in the product backlog. PB14 to PB16 record optional future enhancements. Story identifiers US01 to US13 refer to the separate User Stories and Tasks document and are not sprint numbers.

ID	User story	Priority	When	Est. h
PB01	Project Initiation & System Architecture	**Must**	Week 1	Not recorded
PB02	Core Firmware & Sensor Interfacing	**Must**	Week 2	Not recorded
PB03	HiveMQ Cloud Broker Setup	**Must**	Week 3	Not recorded
PB04	Cellular Communication (SIM7000G)	**Must**	Week 4	Not recorded
PB05	SAEON Dashboard Base Application	**Must**	Week 5	Not recorded
PB06	Power Supply & Solar Integration	**Must**	Week 6	Not recorded
PB07	Advanced UI & Health Diagnostics	**Must**	Week 7	Not recorded
PB08	Hardware Finalization & PCB Soldering	**Must**	Week 8	Not recorded
PB09	Pre-Deployment Auditing & Calibration	**Must**	Week 9	Not recorded
PB10	48-Hour Field Stress Test	**Must**	Week 10	Not recorded
PB11	Presentation Formatting & Content	**Must**	Week 11	Not recorded
PB12	Rehearsal & Submission Prep	**Must**	Week 12	Not recorded
PB13	Project Buffer & Final Review	**Must**	Week 13	Not recorded
PB14	Additional pressure, wind and UV sensing	**Could**	Future	Not recorded

ID	User story	Priority	When	Est. h
PB15	GPS location reporting	**Could**	Future	Not recorded
PB16	Secure over-the-air firmware updates	**Could**	Future	Not recorded

Must means required for the current agreed scope; Could identifies future enhancements. Revised estimates are Not recorded and are not commitments. Field work remains In Progress even though its original planned period has passed.

Sprint 1: Project Initiation & System Architecture

Week 1 · 28 July – 31 July 2026 · PB01 · Tuesday–Friday

Sprint goal Agree the project scope, verify components, set up development tools and document the architecture.

Selected backlog item PB01: Project Initiation & System Architecture. The daily task cards below match the product backlog.

Dependencies Agreed scope, component access and team availability.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
1.01	Day 1: Define Project Scope & Board Setup	Redah	Not recorded	Completed
1.02	Day 2: Component Verification & Sourcing	Redah	Not recorded	Completed
1.03	Day 3: Development Environment Configuration	Qaasim	Not recorded	Completed
1.04	Day 4: Architecture Diagram & Data Flow	Ethan	Not recorded	Completed

Acceptance criteria

- Scope, roles, component choices and architecture are documented.
- Trello and GitHub responsibilities are clear.
- ESP32 WROOM-32 and SHT31 are the selected controller and temperature/humidity sensor.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 2: Core Firmware & Sensor Interfacing

Week 2 · 4 August – 7 August 2026 · PB02 · Tuesday–Friday

Sprint goal Read temperature and rain-gauge tips on the ESP32 and prepare the firmware power behaviour.

Selected backlog item PB02: Core Firmware & Sensor Interfacing. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–1, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
2.01	Day 1: ESP32 Base Setup & Serial Output	Redah	Not recorded	Completed
2.02	Day 2: SHT31 Sensor Logic	Redah	Not recorded	Completed
2.03	Day 3: TR-525M Rain Gauge Logic	Redah	Not recorded	Completed
2.04	Day 4: Firmware Formatting & Low Power Mode	Redah	Not recorded	Completed

Acceptance criteria

- Controlled tips are counted once using a verified conversion factor.
- Temperature readings and invalid-sensor states are identifiable.
- Firmware formatting and low-power behaviour have recorded bench checks.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 3: HiveMQ Cloud Broker Setup

Week 3 · 11 August – 14 August 2026 · PB03 · Tuesday–Friday

Sprint goal Configure HiveMQ permissions, test MQTT delivery and connect Firebase storage.

Selected backlog item PB03: HiveMQ Cloud Broker Setup. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–2, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
3.01	Day 1: Broker Provisioning	Qaasim	Not recorded	Completed
3.02	Day 2: Topic Structuring & Access Control	Qaasim	Not recorded	Completed
3.03	Day 3: Network Connection & Publish Testing	Qaasim	Not recorded	Completed
3.04	Day 4: Database Storage Integration	Qaasim	Not recorded	Completed

Acceptance criteria

- HiveMQ access uses documented publish and subscribe permissions.
- Sample readings pass through the MQTT route securely.
- Firebase stores station-linked, timestamped telemetry.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 4: Cellular Communication (SIM7000G)

Week 4 · 18 August – 21 August 2026 · PB04 · Tuesday–Friday

Sprint goal Send real readings through SIM7000G cellular MQTT and check connection stability.

Selected backlog item PB04: Cellular Communication (SIM7000G). The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–3, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
4.01	Day 1: SIM Module Wiring & Power	Redah	Not recorded	Completed
4.02	Day 2: Network Bearer Configuration	Qaasim	Not recorded	Completed
4.03	Day 3: MQTT over Cellular	Redah	Not recorded	Completed
4.04	Day 4: Stability & Error Handling	Qaasim	Not recorded	Completed

Acceptance criteria

- Real messages reach the dashboard over cellular with Wi-Fi disabled.
- Network registration and modem configuration are recorded.
- Outage recovery and duplicate handling are checked.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 5: SAEON Dashboard Base Application

Week 5 · 25 August – 28 August 2026 · PB05 · Tuesday–Friday

Sprint goal Build the dashboard, live displays, historical filters and the interface for planned cloud commands.

Selected backlog item PB05: SAEON Dashboard Base Application. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–4, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
5.01	Day 1: App Skeleton & Routing	Angelo	Not recorded	Completed
5.02	Day 2: Real-Time Visualization	Angelo	Not recorded	Completed
5.03	Day 3: Historical Data & Filtering	Angelo	Not recorded	Completed
5.04	Day 4: Cloud Command Prep	Qaasim	Not recorded	Completed

Acceptance criteria

- Current readings show units and latest update times.
- Historical filters and data views work against stored records.
- Cloud-command preparation is documented without claiming remote-control acceptance.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 6: Power Supply & Solar Integration

Week 6 · 1 September – 4 September 2026 · PB06 · Tuesday–Friday

Sprint goal Integrate battery and solar charging and verify the alternative power arrangement.

Selected backlog item PB06: Power Supply & Solar Integration. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–5, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
6.01	Day 1: Battery & Charge Controller Setup	Redah	Not recorded	Completed
6.02	Day 2: Solar Panel Integration	Redah	Not recorded	Completed
6.03	Day 3: Mains Power Alternative Testing	Qaasim	Not recorded	Completed
6.04	Day 4: System Power Flow Audit	Qaasim	Not recorded	Completed

Acceptance criteria

- Battery and charging components are integrated and checked.
- Solar and alternative power test results are recorded.
- The power-flow audit records limitations.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 7: Advanced UI & Health Diagnostics

Week 7 · 8 September – 11 September 2026 · PB07 · Tuesday–Friday

Sprint goal Provide station diagnostics, CSV export, responsive views and an end-to-end pipeline check.

Selected backlog item PB07: Advanced UI & Health Diagnostics. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–6, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
7.01	Day 1: Develop Station Health Diagnostics UI Component	Angelo	Not recorded	Completed
7.02	Day 2: Implement CSV Export Functionality	Qaasim	Not recorded	Completed
7.03	Day 3: Mobile & Tablet Responsiveness	Angelo	Not recorded	Completed
7.04	Day 4: Full End-to-End Data Pipeline Ping Test	Qaasim	Not recorded	Completed

Acceptance criteria

- Station health and stale-data states are visible.
- CSV export matches the selected data.
- Responsive views and the complete data path are checked.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 8: Hardware Finalization & PCB Soldering

Week 8 · 15 September – 18 September 2026 · PB08 · Tuesday–Friday

Sprint goal Transfer the prototype to its permanent circuit and mount it in the prepared enclosure.

Selected backlog item PB08: Hardware Finalization & PCB Soldering. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–7, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
8.01	Day 1: Circuit Transfer & Layout	Redah	Not recorded	Completed
8.02	Day 2: PCB Soldering	Redah	Not recorded	Completed
8.03	Day 3: Enclosure Preparation	Redah	Not recorded	Completed
8.04	Day 4: Component Mounting & Sealing	Redah	Not recorded	Completed

Acceptance criteria

- The permanent circuit matches the reviewed wiring.
- Soldering and mounting are checked.
- The enclosure is prepared without treating assembly as field acceptance.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 9: Pre-Deployment Auditing & Calibration

Week 9 · 22 September – 25 September 2026 · PB09 · Tuesday–Friday

Sprint goal Audit power and charging, then proceed with installation and baseline field calibration.

Selected backlog item PB09: Pre-Deployment Auditing & Calibration. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–8, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
9.01	Day 1: Power Drain Audit	Qaasim	Not recorded	Completed
9.02	Day 2: Solar Charging Verification	Redah	Not recorded	Completed
9.03	Day 3: Full System Deployment	Redah	Not recorded	In Progress
9.04	Day 4: Baseline Calibration	Redah	Not recorded	In Progress

Acceptance criteria

- Power drain and solar charging checks are Completed.
- Installation and baseline field calibration remain In Progress.
- Final site permission, protection and field records are required before acceptance.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 10: 48-Hour Field Stress Test

Week 10 · 29 September – 2 October 2026 · PB10 · Tuesday–Friday

Sprint goal Complete and audit the 48-hour field stress test; this field-validation work remains In Progress.

Selected backlog item PB10: 48-Hour Field Stress Test. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–9, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
10.01	Day 1: Stress Test Initiation	Qaasim	Not recorded	In Progress
10.02	Day 2: Active Monitoring	Qaasim	Not recorded	In Progress
10.03	Day 3: Manual Field Measurements	Ethan	Not recorded	In Progress
10.04	Day 4: Data Auditing & Conclusion	Ethan	Not recorded	In Progress

Acceptance criteria

- The test duration, uptime and delivery gaps are recorded when available.
- Reference measurements and timestamps can be compared.
- Field conclusions are not marked Completed before the test and audit finish.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 11: Presentation Formatting & Content

Week 11 · 6 October – 9 October 2026 · PB11 · Tuesday–Friday

Sprint goal Complete presentation content, diagrams, dashboard evidence and technical documentation.

Selected backlog item PB11: Presentation Formatting & Content. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–10, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
11.01	Day 1: Presentation Outlining	Ethan	Not recorded	Completed
11.02	Day 2: Technical Diagrams & Architecture Slides	Ethan	Not recorded	Completed
11.03	Day 3: Dashboard & Software Slides	Ethan	Not recorded	Completed
11.04	Day 4: Technical Documentation	Ethan	Not recorded	Completed

Acceptance criteria

- Slides describe the actual architecture and current results.
- The setup guide and technical documentation are complete for the prototype.
- Open field-validation limitations are explicit.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 12: Rehearsal & Submission Prep

Week 12 · 13 October – 16 October 2026 · PB12 · Tuesday–Friday

Sprint goal Finalise the slides, rehearse the demonstration and compile the prototype submission materials.

Selected backlog item PB12: Rehearsal & Submission Prep. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–11, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
12.01	Day 1: Slide Deck Finalization	Ethan	Not recorded	Completed
12.02	Day 2: Group Rehearsal 1	Rameez	Not recorded	Completed
12.03	Day 3: Timed Rehearsal & Feedback	Rameez	Not recorded	Completed
12.04	Day 4: Final Compilation	Rameez	Not recorded	Completed

Acceptance criteria

- Speaking parts and rehearsals are prepared.
- A backup demonstration and submission package are compiled.
- Preparation does not imply a submission receipt or final field approval.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Sprint 13: Project Buffer & Final Review

Week 13 · 20 October – 23 October 2026 · PB13 · Tuesday–Friday

Sprint goal Review the prototype documentation package and record outstanding deployment work separately.

Selected backlog item PB13: Project Buffer & Final Review. The daily task cards below match the product backlog.

Dependencies Relevant outputs from Sprints 1–12, team capacity and required equipment or services.

Sprint tasks

Task ID	Task and expected output	Lead	Est. h	Status
13.01	Day 1-4: Buffer & Prototype Documentation Sign-off	Redah	Not recorded	Completed

Acceptance criteria

- The prototype documentation and presentation package are reviewed.
- The deployment and field-validation actions remain open.
- Prototype sign-off is distinguished from field-deployment acceptance.

Review evidence Linked task outputs, configuration or wiring records, screenshots, logs and review notes as applicable. No deployment acceptance or submission receipt is inferred.

Review date: Not recorded _____ _____ _____	
Review notes: Record actual evidence and feedback; field work remains open where marked In Progress.	**Retrospective: Record the team's actual improvement actions; no dated meeting outcome supplied.**

Release and handover

Release review checklist

- Show real rain tips and temperature readings arriving on the live dashboard through cellular MQTT.
- Show bench calibration, recorded test results, message-delivery evidence and outage handling. Clearly separate these from the In Progress baseline field calibration and 48-hour validation.
- Confirm that the tagged code, wiring guide and setup instructions rebuild the prototype that was assessed.
- Record the known limitations, the extras we postponed and the feedback from the lecturer or SAEON.
- Prototype documentation sign-off is Completed. Final field-deployment acceptance remains open until installation, baseline calibration and field validation are complete. Retain an actual submission receipt only when a submission is made.

Supporting project documents

Agile Team Charter SAEON Rainfall Monitoring: team roles, 13-week milestones, working agreements and Definition of Done.

SAEON ESP32 Rainfall Monitoring Basic Presentation. Where we got the problem, who benefits, the core hardware and the cellular MQTT data flow.

SAEON Project Sprint Plan. Our reference for task ideas, sensor integration, communication tests and handover activities.